

Shakiba Ahmadimehr

Biomedical Engineer & Data Analyst

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RESEARCH INTERESTS

- **Biomedical Image & Signal Processing:** Task-based / Resting-state Brain mapping, Neuroimaging using MRI, DTI, fMRI
- **Computational Neuroscience**
- **Biomedical Pattern Recognition**
- **Machine Learning with Medical Approach**
- **Cognitive Neuroscience:** Studying human brain functional organization with network-based approaches in Cognition, Language, Learning & Memory

EDUCATION

University of California, Irvine Irvine, CA
Ph.D. Student in Biomedical Engineering Sep 2023 – Jun 2025

- Completed doctoral-level coursework and research rotations focusing on neuroimaging and machine learning models.

Master of Science (M.Sc.) in Biomedical Engineering Sep 2019 – Jan 2022

- **Specialization:** Neuroimaging, Brain Decoding & Machine Learning
- Focused on computational modeling of brain data, advanced signal analytics, and machine learning techniques applied to medical diagnostics.

Bachelor of Engineering (B.E.) in Biomedical Engineering Sep 2015 – Feb 2019

- **Core Foundations:** Coursework in biomedical instrumentation, signals and systems, biostatistics, and foundational programming.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, Bash, Command Line, Git

Neuroimaging & Analysis: fMRI, DTI, Network Analysis, Brain Connectivity, Time Series Analysis, Surface Analysis

Frameworks & Tools: PyTorch, NumPy, Pandas, Scikit-Learn, Streamlit, Jupyter, MRtrix, FSL, FreeSurfer, BIDS

Statistics & Machine Learning: Linear Models, Mixed Models, Prediction, Classification, Clustering

Languages: English, Persian

RESEARCH & WORK EXPERIENCE

Graduate Researcher Sep 2024 – Jun 2025
Sanger Lab (Prof. Terence David Sanger) – UC Irvine

- **Neuroimaging & Tractography:** Processed structural MRI/DTI datasets, utilizing FreeSurfer to parcellate cortical and subcortical regions and mapping contralateral fiber connections between targeted brain areas using MRtrix.

Graduate Research Rotational Student Jan 2024 – Jul 2024
Visual Perception & Neuroimaging Lab (VPNL) (Prof. Emily Grossman) – UC Irvine

- Completed a 6-month lab rotation (Winter and Spring quarters) in the Visual Perception & Neuroimaging Lab (VPNL), focusing on human subject fMRI methodologies, signal preprocessing, and pattern classification.

Graduate Research Student Sep 2023 – Sep 2024
Biomedical Engineering Department – UC Irvine

- Conducted interdisciplinary research rotations across departmental units.

Teaching Assistant Oct 2024 – Jun 2025
Samueli School of Engineering – UC Irvine

- Evaluated coursework and provided constructive feedback to support academic development in engineering subjects.

Research Assistant

Aug 2022 – May 2023

Multiscale Bioimaging & Bioinformatics (MBB) Lab – Tulane University

- Conducted computational research supervised by Prof. Yu-Ping Wang.
- Designed machine learning and deep learning models to classify complex fMRI time-series data.

Teaching Assistant

Feb 2020 – Jul 2020

Graduate Level Courses: Biological Systems Modeling & Biological Signal Processing

- Provided weekly assignments and summaries of core course concepts.
- Received student assignments and projects, graded coursework, and provided constructive feedback.

RESEARCH PROJECTS

Structural Connectivity Analysis*DTI & Graph Theory*

- Quantified structural brain networks using Diffusion Tensor Imaging pipelines to analyze topological organization and network connectivity metrics.

Unsupervised Pattern Discovery*Machine Learning & fMRI*

- Applied clustering algorithms and unsupervised latent variable learning to uncover hidden functional brain states across multi-subject fMRI datasets.

fMRI Signal Modeling & Brain Decoding*FIR Modeling & Classification*

- Reconstructed temporal fMRI dynamics using Finite Impulse Response (FIR) modeling and classified cognitive states during cued and uncued tasks.

Evaluation of Brain Connectivity in Parkinson's Disease*Master's Thesis Research*

- Investigated resting-state functional connectivity via Independent Component Analysis (ICA) and Granger Causality Analysis (GCA) mapped to the AAL atlas.

Facial Attractiveness Classification*Bachelor's Senior Project*

- Implemented end-to-end machine learning pipelines incorporating data preprocessing, geometric feature extraction, statistical analysis, and classification models to identify facial attractiveness.

PUBLICATIONS & PREPRINTS

- **Ahmadimehr, S.** et al. (2022). "Investigation of linear & non-linear connectivity." *Research Preprint*. [Link to Paper]
- **Ahmadimehr, S.** et al. (2022). "Alteration of resting-state brain networks in Parkinson's disease: A multi-aspect study of functional and effective connectivity." *Research Preprint*. [Link to Paper]
- **Ahmadimehr, S.**, & Moridani, M. K. (2021). "Identify Attractive and Unattractive Individuals Based on Geometric Features Using Neural Network." *Traitement du Signal*. [Link to Paper]
- **Ahmadimehr, S.**, & Moridani, M. K. (2020). "Evaluating facial attractiveness through proportions analysis based on geometric features." *Journal of Image Processing & Processes*. [Link to Paper]